

Course Title	Introduction to the Internet of Things	Course Code	IO3031
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering in Society (WK7)	Credit Hrs.	3+1
Knowledge Area	Computer Engineering (KA01)	Grading Scheme	Relative
HEC Knowledge Area	Depth Electives	Applicable From	Spring 2023
SDG	11 Sustainable Cities and Communities	PBL	1
Pre-requisite(s)	-		

Course Objective	To comprehend the Internet of Things and explore in detail its ecosystem/architecture, protocol stack, enabling technologies such as embedded systems, hardware and software, wireless connectivity standards, the underlying Internet, its applications and challenges. To develop a simple prototype application in order to build foundation for further development.
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No.	Assigned Program Learning Outcome (PLO)
01	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
06	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
07	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

A = Assignment, Q = Quiz, M = Midterm, F=Final

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Use understanding of IoT architecture and identify practical application areas.	Q1, M1, F	C3	1
2	Apply the knowledge of Internet Connectivity Principles, IoT Networking Technologies, Protocols, and standards.	A1, Q2, M2, F	C3	1
3	Explore IoT applications based on hardware, cloud storage and data analysis for industrial and societal usage.	A2, Q3 F	C4	6
4	Evaluate IoT solutions in terms of their environmental impact and contribution to sustainable development.	A3, F	C5	7

Text Book(s)	Title	Internet of Things: A Hands-on Approach
	Author	Arshdeep Bahga and Vijay Madisetti
	Publisher	University Press
Ref. Book(s)	Title	Interconnecting Smart Objects with IP: The Next Internet
	Author	Jean-Philippe Vasseur and Adam Dunkels
	Publisher	Morgan Kaufmann
	Title	IoT Fundamentals
	Author	David Hanes, Gonzalo Salgueiro, Patrick Grossetet
	Publisher	Cisco Press

Week	Course Contents/Topics	Chapter*	CLO*
1	Module – 1: IoT Overview, System Architecture and Design Principles - Course Overview (CLO1) - Introduction to IoT - Related areas and terminologies - IoT vision and growth - Modern IoT trends - IoT Platforms - IoT Connectivity - Issues and Challenges	1	1
2-3	- Characteristics of IoT - Physical Design of IoT - Logical Design of IoT - IoT Protocols - IoT Levels & Deployment Templates - Setting up Python - Introduction to Python	6	1
4-5	- SDN architecture: Control plane improvements with SDN - Network automation, open stack, ONOS SDN controllers - Perception layer of IoT - Components of an IoT Device - Sensors and Actuators		2
6-7	Design Principles for the Web Connectivity for Connected-Devices - IoT Design Methodology - Purpose and Requirements Specification - Process Specification	5	2

	• Domain Model Specification • Information Model Specification • Service level specification • IoT level specification • Functional view specification • Operational view specification • Device and component integration • Application development		
8-9	Module – 2: TCP/IP Internet Model and IoT Networking Internet (TCP/IP) Connectivity Principles IoT Networking: • Communication layers and protocols in IoT • Communication principles • IoT standard protocols • Phy+Link Layer (LR WPAN, Bluetooth Low Energy, WiFi, ZigBee, LoRA, Cellular) • Network (6LoWPAN, LoRAWAN) • Transport (TCP/UDP) • Application (CoAP, MQTT)		2
10-11	LPWAN • Introduction • Types of LPWAN • Characteristics • LoRa-The DNA of IoT • LoRA Technology • Three classes of end devices • LoRaWAN Performance		2
12	Module – 3: Developing Internet of Things • Basic building blocks of an IoT Device • IoT Physical Devices & Endpoints • Prototyping the Embedded Devices for IoTs • IoT Operating Systems • Exemplary device: Raspberry Pi • Raspberry Pi Interfaces • Programming Raspberry Pi with Python • Other IoT Devices	7	3

13	Module – 4: Building IoT with Cloud, Data Collection and Analysis • Introduction to Cloud Computing • Cloud Architecture and characteristics • Cloud service models • Study of various cloud offering • Opportunities and Challenges in cloud computing	8	3
14	• Sensing components and devices • Sensor modules, nodes • Motes and systems.		2
15	• Wireless technologies for the IoT • Edge connectivity and protocols • Wireless sensor networks		3

Assessment Tools	Weightage
Quizzes (3)	10.0%
Assignment (3)	10.0%
Midterm (I+II)	15.0% + 15.0%
Final Exam	50.0%